

[illegible]



INSTITUTE OF ENGINEERING & TECHNOLOGY

MOHANLAL SUKHADIA UNIVERSITY, UDAIPUR

Name : Lavina Tamboli | Class: Btech-CSE (3 sem) | Subject: SE Lab

Program -1

AIM: WAP a program to print "Hello World".

Program:

```
public class helloWorld
{
    Run main | Debug main | Run | Debug
    public static void main(String[] args)
    {
        System.out.println(x: "hello world");
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac helloWorld.java } ; if ($?) { java helloWorld }
hello world
PS L:\C++\java>
```



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Program -2

AIM: WAP a Program to print two statements int two new lines.

Program:

```
public class TwoStatements {  
    Run main | Debug main | Run | Debug  
    public static void main(String[] args) {  
        System.out.println(x: "First statement");  
        System.out.println(x: "Second statement");  
    }  
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac TwoStatements.java } ; if ($?) { java TwoStatements }  
First statement  
Second statement  
PS L:\C++\java>
```



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Program -3

AIM: WAP a Program to add two numbers using constant values.

Program:

```
public class AddNumber {  
    Run main | Debug main | Run | Debug  
    public static void main(String[] args) {  
        int a=20;  
        int b=30;  
        System.out.println("First number="+ a);  
        System.out.println("Second number="+b);  
        int sum = a+b;  
        System.out.println("Sum = "+sum);  
    }  
}
```

Output:

```
PS L:\C++> cd "L:\C++\java\" ; if ($?) { javac AddNumber.java } ; if ($?) { java AddNumber }  
First number=20  
Second number=30  
Sum = 50  
PS L:\C++\java>
```



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Program-4

AIM: WAP to add two integer numbers entered by the user using scanner class.

Program:

```
import java.util.Scanner;
public class AddTwoNumbers {
    Run main | Debug main | Run | Debug
    public static void main(String[] args){
        Scanner sc= new Scanner(System.in);
        System.out.println(x: "Enter first number:");
        int num1= sc.nextInt();
        System.out.println(x: "Enter second number:");
        int num2= sc.nextInt();
        int sum = num1+num2;
        System.out.println("Sum="+sum);
        sc.close();
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac AddTwoNumbers.java } ; if ($?) { java AddTwoNumbers }
Enter first number:
40
Enter second number:
50
Sum=90
PS L:\C++\java>
```



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Program-5

AIM: WAP to add two float numbers entered by the user using scanner class.

Program:

```
import java.util.Scanner;
public class AddTwoFloatNumbers {
    Run main | Debug main | Run | Debug
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in); //object creation of scanner
        System.out.println(x: "Enter first number");
        float num1 =sc.nextFloat(); //user input float value
        System.out.println(x: "Enter second number");
        float num2=sc.nextFloat();
        Float sum = num1+num2;
        System.out.println("sum="+sum);
        sc.close();
    }
}
```

Output:

```
PS L:\C++> cd "I:\C++\java\" ; if ($?) { javac AddTwoFloatNumbers.java } ; if ($?) { java AddTwoFloatNumbers }
Enter first number
20.4
Enter second number
50.67
sum=71.07
PS L:\C++\java>
```



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Program-6

AIM: Write a program to create a simple calculator that can perform basic arithmetic operations(Addition, Subtraction, Multiplication , Division) on two numbers.

Program:

```
import java.util.Scanner;
public class Calculator {
    Run main | Debug main | Run | Debug
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter first number:");
        int num1= sc.nextInt();
        System.out.println(x: "Enter second number:");
        int num2= sc.nextInt();
        System.out.println(x: "Enter your choice:1.(+),2.(-),3.(*),4.(/)");
        char choice=sc.next().charAt(index: 0);

        switch(choice)
        {
            case '+':
                int sum =num1+num2;
                System.out.println("sum="+sum);
                break;
            case '-':
                int sub = num1-num2;
                System.out.println("sub="+sub);
                break;
            case '*':
                int multi = num1*num2;
                System.out.println("multi =" +multi);
                break;
            case '/':
                int divi = num1/num2;
                System.out.println("divi="+divi);
                break;
        }
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac Calculator.java } ; if ($?) { java Calculator }
Enter first number:
20
Enter second number:
40
Enter your choice:1.(+),2.(-),3.(*),4.(/)
*
multi =800
PS L:\C++\java>
```



Program-7

AIM: Write a java program to swap two numbers using third variable.

Program:

```
public class SwapTwoNumbers {  
    Run main | Debug main | Run | Debug  
    public static void main(String[] args)  
    {  
        int num1=10;  
        int num2=20;  
        int temp;  
        System.out.println(x: "Before Swapping");  
        System.out.println("first number:"+num1);  
        System.out.println("second number:"+num2);  
        temp=num1;  
        num1=num2;  
        num2=temp;  
        System.out.println(x: "After swapping:");  
        System.out.println("first number:"+num1);  
        System.out.println("Second number"+num2);  
    }  
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac SwapTwoNumbers.java } ; if ($?) { java SwapTwoNumbers }  
Before Swapping  
first number:10  
second number:20  
After swapping:  
first number:20  
Second number10  
PS L:\C++\java>
```



Program-8

AIM: Write a java program to swap two numbers without using third variable.

Program:

```
public class SwapWithoutThirdVariable {  
    Run main | Debug main | Run | Debug  
    public static void main(String[] args)  
    {  
        int a=10;  
        int b=20;  
        System.out.println(x: "Before Swapping");  
        System.out.println("first number:"+a);  
        System.out.println("second number:"+b);  
        a = a+b;  
        b = a-b;  
        a = a-b;  
        System.out.println(x: "After swapping:");  
        System.out.println("first number:"+a);  
        System.out.println("Second number"+b);  
    }  
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac SwapWithoutThirdVariable.java } ; if ($?) { java SwapWithoutThirdVariable }  
Before Swapping  
first number:10  
second number:20  
After swapping:  
first number:20  
Second number10  
PS L:\C++\java>
```



Program-9

AIM: write a java program to find the factorial of a number.

Program:

```
import java.util.Scanner;
public class Factorial {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc= new Scanner(System.in);
        System.out.println(x: "Enter the number:");
        int n = sc.nextInt();
        int fact =1;
        for (int x = n; x >=1; x--) {
            fact = x*fact;
        }
        System.out.println("Factorial is:"+fact);
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac Factorial.java } ; if ($?) { java Factorial }
Enter the number:
5
Factorial is:120
PS L:\C++\java>
```



Program -10

AIM: Write a java program to find the Fibonacci series.

Program:

```
import java.util.Scanner;
public class fabonacci {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc= new Scanner(System.in);
        System.out.println(x: "Enter the number:");
        int n = sc.nextInt();
        int a=0;
        int b=1;
        int c;
        System.out.println(x: "Fabonacci series:");
        for (int i = 0; i < n; i++) {
            System.out.println(" "+a);
            c = a+b;
            a=b;
            b=c;
        }
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac fabonacci.java } ; if ($?) { java fabonacci }
Enter the number:
10
Fabonacci series:
0
1
1
2
3
5
8
13
21
34
PS L:\C++\java>
```



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Experiment – 11

AIM: WAP to find whether a given number is a leap year or not

Program:

```
import java.util.Scanner;
public class leapYear {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter your year:");
        int year= sc.nextInt();
        if((year%4==0)&&(year%100!=0) || year%400==0)
        {
            System.out.println(x: "this year is a leap year");
        }
        else
            System.out.println(x: "this year is not a leap year");
    }
}
```

Output:

```
PS L:\C++> cd "L:\C++\java\" ; if ($?) { javac leapYear.java } ; if ($?) { java leapYear }
```

```
Enter your year:
```

```
2024
```

```
this year is a leap year
```

```
Enter your year:
```

```
2025
```

```
this year is not a leap year
```



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Experiment -12

AIM: WAP to find reverse of a number

Program:

```
import java.util.Scanner;
public class ReverseNumber {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter a number");
        int number = sc.nextInt();
        System.out.println(x: "Reverse number:");
        while(number>0){
            int digit = number%10;
            number = number/10;
            System.out.print(digit);
        }
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac ReverseNumber.java } ; if ($?) { java ReverseNumber }
Enter a number
4923
Reverse number:
3294
```



Experiment-13

AIM: WAP to find whether given number is palindrome or not.

Program:

```
import java.util.Scanner;
public class PalindromeNumber {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter a number");
        int number = sc.nextInt();
        int num = number;
        int reverse=0;
        while(number>0){
            int digit = number%10;
            reverse = (reverse*10)+digit;
            number = number/10;
        }
        if(num == reverse )
        {
            System.out.println(x: "it is a palindrome number");
        }
        else
        {
            System.out.println(x: "it is not palindrome number");
        }
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac PalindromeNumber.java } ; if ($?) { java PalindromeNumber }
Enter a number
12321
it is a palindrome number

Enter a number
12452
it is not palindrome number
```



Experiment-14

AIM: WAP to find whether given number is Armstrong or not

Program:

```
import java.util.Scanner;
public class ArmstrongNumber {

    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter a number");
        int number = sc.nextInt();
        int OriginalNumber = number;
        int result=0;
        int n=0;
        while(number>0)
        {
            number= number/10;
            n++;
        }
        number= OriginalNumber;
        while(number>0){
            int digit = number%10;
            result += Math.pow(digit,n);
            number = number/10;
        }

        if(OriginalNumber == result )
        {
            System.out.println(x: "it is a Armstrong number");
        }
        else
        {
            System.out.println(x: "it is not Armstrong number");
        }
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac ArmstrongNumber.java } ; if ($?) { java ArmstrongNumber }
Enter a number
153
it is a Armstrong number

Enter a number
1634
it is a Armstrong number
```



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Experiment-15

AIM: WAP to find Square root of a number using Math class.

Program:

```
import java.util.Scanner;
public class SquareRoot {
    Run | Debug | Run main | Debug main
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a number:");
        int number = sc.nextInt();
        double root = Math.sqrt(number);

        System.out.println("Root of a given number is:" + root);
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac SquareRoot.java } ; if ($?) { java SquareRoot }
Enter a number:
144
Root of a given number is:12.0
```



Experiment -16

AIM: WAP to generate star polynomial for given user input of rows.

Program:

```
import java.util.Scanner;
public class Pyramid {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter number of rows:");
        int rows = sc.nextInt();
        for(int i=0; i<rows; i++){
            for(int j=0; j<rows-i; j++){
                System.out.print(s: " ");
            }
            for(int k=0; k<=i;k++) {
                System.out.print(s: "*");
                System.out.print(s: " ");
            }
            System.out.println(x: "\n");
        }
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac Pyramid.java } ; if ($?) { java Pyramid }
Enter number of rows:
5
    *
  * *
 * * *
* * * *
* * * * *
```



Experiment-17

AIM: WAP to print an array in reverse order

Program:

```
import java.util.Scanner;
public class ReverseAnArray {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter length of array:");
        int length = sc.nextInt();
        System.out.println("Enter "+length+" elements of array:");
        int[] arr= new int[length];
        for (int i=0;i<length; i++)
        {
            arr[i] = sc.nextInt();
            // System.out.println("Array element are:"+ arr[i]);
        }
        System.out.println(x: "Reverse an Array:");
        for(int i = length-1; i>=0; i--){
            System.out.print(arr[i]+" ");
        }
    }
}
```

Output:

```
PS L:\C++> cd "I:\C++\java\" ; if ($?) { javac ReverseAnArray.java } ; if ($?) { java ReverseAnArray }
Enter length of array:
5
Enter 5 elements of array:
10
20
30
40
50
Reverse an Array:
50 40 30 20 10
```



Experiment – 18

AIM: WAP to print odd number of an Array.

Program:

```
// WAP to print odd number of an array
import java.util.Scanner;
public class oddnumberArray {
    Run | Debug
    public static void main(String[] args)
    {
        try (Scanner sc = new Scanner(System.in)){
            System.out.println("enter length of array: ");
            int length = sc.nextInt();
            int[] array = new int[length];
            System.out.println("enter " +length+ " elements of array: ");
            for(int i = 0;i < length;i++)
            {
                array[i] = sc.nextInt();
            }
            System.out.println("odd position of an Array: ");
            for(int i=1; i<length;i+=2){
                System.out.print(array[i]+" ");
            }
        }
    }
}
```

Output:

```
ate/var/folders/rv/57y3t45d1716xfpgdw_wj8_c0000gn/T/vscodesws_66937/jdt_ws/jdt.ls-java-project/bin oddnumberArray
enter length of array:
5
enter 5 elements of array:
10
20
30
40
50
odd position of an Array:
20 40
```



Experiment-19

AIM: WAP to print smallest elements of an array.

Program:

```
import java.util.Scanner;
public class smallestelementarray {
    Run | Debug
    public static void main(String[] args){
    try (Scanner sc = new Scanner(System.in)){
        System.out.println("enter length of array: ");
        int length = sc.nextInt();
        int[] array = new int[length];
        System.out.println("enter " +length+ " elements of array: ");
        for(int i = 0; i < length; i++)
        {
            array[i] = sc.nextInt();
        }
        int min = array[0];
        for(int i = 1; i < length; i++){
            if (array[i] < min) {
                min = array[i];
            }
        }
        System.out.println("smallest element of an Array: "+min);
    }
    }
```

Output:

```
va_db35bd85/bin smallestelementarray
enter length of array:
5
enter 5 elements of array:
10
20
30
40
50
smallest element of an Array: 10
```



Experiment-20

AIM: WAP to find the transpose of a given matrix.

Program:

```
import java.util.Scanner;
public class TransposeMatrix {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter number of rows:");
        int rows = sc.nextInt();
        System.out.println(x: "Enter number of columns:");
        int columns = sc.nextInt();
        System.out.println(x: "Enter Elements of matrix:");
        int[][] arr = new int[rows][columns];

        for(int i=0; i<rows; i++){
            for(int j=0; j<columns; j++){
                arr[i][j] = sc.nextInt();
            }
        }
        System.out.println(x: "Original matrix:");
        for(int i=0; i<rows; i++){
            for(int j=0; j<columns; j++){
                System.out.print(arr[i][j]+" ");
            }
            System.out.println(x: " ");
        }

        System.out.println(x: "Transpose of a matrix:");
        for(int i=0; i<columns; i++){
            for(int j=0; j<rows; j++){
                System.out.print(arr[j][i]+" ");
            }
            System.out.println(x: " ");
        }
    }
}
```

Output:

```
PS L:\C++> cd "I:\C++\java\" ; if ($?) { javac Transpose.java } ; if ($?) { java Transpose }
Enter number of rows:
2
Enter number of columns:
3
Enter Elements of matrix:
10
20
30
40
50
60
Original matrix:
10 20 30
40 50 60
Transpose of a matrix:
10 40
20 50
30 60
```



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Experiment-21

AIM: WAP to find whether given matrix is identity or not.

Program:

```
import java.util.Scanner;
public class IdentityMatrix {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter number of rows:");
        int rows = sc.nextInt();
        System.out.println(x: "Enter number of columns:");
        int columns = sc.nextInt();
        System.out.println(x: "Enter Elements of matrix:");
        int[][] arr = new int[rows][columns];

        for(int i=0; i<rows; i++){
            for(int j=0; j<columns; j++){
                arr[i][j] = sc.nextInt();
            }
        }

        int identity=0;
        for(int i=0; i<rows; i++){
            for (int j=0; j<columns; j++){
                if(i==j){
                    if (arr[i][j]==1){
                        identity=1;
                    }
                }
            }
        }

        for(int i=0; i<rows; i++){
            for (int j=0; j<columns; j++){
                if(i!=j){
                    if (arr[i][j]==0){
                        identity=1;
                    }
                }
            }
        }

        if(identity ==1){
            System.out.println(x: "given Matrix is an Identity matrix:");
        }
        else {
            System.out.println(x: "not an identity matrix:");
        }
    }
}
```

Output:

```
PS L:\C++> cd "I:\C++\java\" ; if ($?) { javac IdentityMatrix.java } ; if ($?) { java IdentityMatrix }
Enter number of rows:
3
Enter number of columns:
3
Enter Elements of matrix:
1
0
0
0
1
0
0
0
1
given Matrix is an Identity matrix:
```



Experiment -22

AIM: WAP to get sum of each row and column of matrix.

Program:

```
import java.util.Scanner;
public class SumOfRowsCloumns {
    Run main | Debug main | Run | Debug
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println(x: "Enter number of rows:");
        int rows = sc.nextInt();
        System.out.println(x: "Enter number of columns:");
        int columns = sc.nextInt();
        System.out.println(x: "Enter Elements of matrix:");
        int[][] arr = new int[rows][columns];

        for(int i=0; i<rows; i++){
            for(int j=0; j<columns;j++){
                arr[i][j] = sc.nextInt();
            }
        }

        for (int i = 0; i < rows; i++) {
            int sum = 0;

            for (int j = 0; j < columns; j++) {
                sum += arr[i][j];
            }
            System.out.println("Sum of row " + (i + 1) + " = " + sum);
        }

        for (int j = 0; j < columns; j++) {
            int sum = 0;

            for (int i = 0; i < rows; i++) {
                sum += arr[i][j];
            }
            System.out.println("Sum of column " + (j + 1) + " = " + sum);
        }
    }
}
```

Output:

```
PS L:\C++> cd "l:\C++\java\" ; if ($?) { javac SumOfRowsCloumns.java } ; if ($?) { java SumOfRowsCloumns }
Enter number of rows:
2
Enter number of columns:
3
Enter Elements of matrix:
10
20
30
40
50
60
Sum of row 1 = 60
Sum of row 2 = 150
Sum of column 1 = 50
Sum of column 2 = 70
Sum of column 3 = 90
```



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Experiment-23

AIM: WAP to implement the consttruct overloading

program:

```
import java.util.Scanner;
public class OverloadingExample {

    static int add(int a, int b) {
        return a + b;
    }
    static int add(int a, int b, int c) {
        return a + b + c;
    }
    static double add(double a, double b) {
        return a + b;
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print(s: "Enter first number: ");
        int a = sc.nextInt();

        System.out.print(s: "Enter second number: ");
        int b = sc.nextInt();

        System.out.print(s: "Enter third number: ");
        int c = sc.nextInt();

        System.out.print(s: "Enter first decimal number: ");
        double d1 = sc.nextDouble();

        System.out.print(s: "Enter second decimal number: ");
        double d2 = sc.nextDouble();

        System.out.println("Sum of two numbers: " + add(a, b));
        System.out.println("Sum of three numbers: " + add(a, b, c));
        System.out.println("Sum of two decimals: " + add(d1, d2));
        sc.close();
    }
}
```

Output:

```
PS C:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING> cd "c:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING\" ; if ($?) { javac OverloadingExample.java } ; if ($?) { java OverloadingExample }
Enter first number: 8
Enter second number: 5
Enter third number: 6
Enter first decimal number: 6.9
Enter second decimal number: 9.6
Sum of two numbers: 13
Sum of three numbers: 19
Sum of two decimals: 16.5
PS C:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING>
```



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Experiment-24

AIM: WAP to perform inheritance

program:

```
import java.util.Scanner;

class Animal {
    void sound() {
        System.out.println(x: "Animal makes a sound");
    }
}

class Dog extends Animal {
    void sound() {
        System.out.println(x: "Dog barks");
    }
}

public class InheritanceExample {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.println(x: "Choose animal: 1 for Animal, 2 for Dog");
        int choice = sc.nextInt();

        if (choice == 1) {
            Animal a = new Animal();
            a.sound();
        } else if (choice == 2) {
            Dog d = new Dog();
            d.sound();
        } else {
            System.out.println(x: "Invalid choice");
        }

        sc.close();
    }
}
```

Output:

```
PS C:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING> cd "c:\Users\Himank\OneDrive\
) { java InheritanceExample }
Choose animal: 1 for Animal, 2 for Dog
1
Animal makes a sound
PS C:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING> █
```



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Experiment -25

AIM:WAP to perform try , catch and finally block

Program:

```
import java.util.Scanner;

public class TryCatchExample {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        try {
            System.out.print(s: "Enter first number: ");
            int a = sc.nextInt();

            System.out.print(s: "Enter second number: ");
            int b = sc.nextInt();

            int result = a / b;
            System.out.println("Result: " + result);
        } catch (ArithmeticException e) {
            System.out.println(x: "Error: Cannot divide by zero");
        } catch (Exception e) {
            System.out.println(x: "Error: Please enter valid numbers");
        } finally {
            System.out.println(x: "Finally block executed");
            sc.close();
        }
    }
}
```

Output:

```
PS C:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING> cd "c:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING\" ; if
java TryCatchExample }
Enter first number: 5
Enter second number: 5
Result: 1
Finally block executed
PS C:\Users\Himank\OneDrive\Desktop\SEM 4\PYTHON_PROGRAMMING> █
```